TraceSOC Executive Brief

From fragmented security evidence to defensible, reproducible conclusions

> TraceSOC is a provenance-first SOC/DFIR evidence-reconstruction engine. Its central promise is not a dramatic verdict; it is that consequential analytical claims remain recoverable to the exact raw evidence that supports them, while uncertainty and unsupported claims remain visible.

Project: SOC-A5
Public variant: V3
Candidate: UBI-2026-0091
Evidence marker: UBI-A9-4BC183701F3C
Frozen technical commit: 8751a679f36b4983cdd3d2b0b35f44e9a11c4f39

Submission note: This report is evidence-bounded. It distinguishes measured capability from unmeasured accuracy, causality, maliciousness, live response execution, and ROI.

Problem, Impact, Proof, Defense

UBI asks four questions at the start of the finalist story. TraceSOC can answer all four directly, and each answer is tied to measured evidence rather than marketing claims.

Problem - What real security problem are you solving, and for whom?

SOC analysts and DFIR investigators work across fragmented email, endpoint, packet and network evidence. As investigations progress, conclusions can become detached from the exact observations that support them. That weakens reproducibility, analyst handoff, auditability and later technical review. TraceSOC is for SOC analysts, incident responders, DFIR practitioners, and the reviewers or security teams that need to validate how a conclusion was reached.

Impact - What becomes safer, faster, clearer, or more resilient because your project exists?

TraceSOC makes investigations clearer and more defensible, safer from unsupported analytical certainty, and more resilient to handoff and later review by preserving exact evidence lineage and explicit uncertainty. Deterministic reproduction improves repeatability. Analyst-time reduction was not measured, so no claim that investigations are faster is made.

Proof - How will a judge independently verify that it works?

The judge can verify the pre-holdout technical freeze; 5,442/5,442 raw-locator coverage in the controlled six-source case; one complete response-to-original-SHA reverse trace; 23,145/23,145 external Zeek reconciliation; 156,458/156,458 sealed-holdout reconciliation with 938,748 detector evaluations and no post-freeze tuning; 524/524 regression with 4/4 public acceptance; two fresh reproductions with the same 11 core hashes; and the measured 198,463-record resource benchmark.

Defense - Can you explain every major technical choice, limitation, and trade-off?

Yes. The architecture deliberately treats provenance as part of the event contract, preserves unsupported evidence, separates observation from inference, uses representation-specific adapters, bounds correlation to what evidence supports, aggregates downstream reasoning without deleting packet multiplicity, freezes before holdout evaluation, distinguishes recommendations from executed response, and records open limitations instead of inventing accuracy, causality, maliciousness, ROI or cross-publisher generalization.

Why TraceSOC had to exist

The project began from a practical SOC/DFIR problem: security investigations are multi-source, but analytical narratives often become easier to read than they are to verify.

Fragmentation is only the first problem

An investigation can contain email content, packet captures, Zeek network records, Sysmon process and network events, PowerShell Operational logs, and Windows Security events. Each source has its own representation, identifiers, timing semantics and failure modes. A useful system must normalize them without erasing where each observation came from.

The deeper problem is evidence-to-conclusion detachment

Once evidence is normalized, detection and correlation create another risk: the downstream conclusion can become detached from the exact source observation. A finding may sound convincing while the reviewer cannot recover the supporting raw record, determine which part is observed versus inferred, or see what the relationship does not prove.

> **Design response:** TraceSOC therefore treats raw recoverability, explicit uncertainty and bounded inference as product requirements - not as documentation added after analysis.

Who benefits

- SOC analysts who need to move from alert-like signals to evidence-backed investigation findings.
- Incident responders and DFIR practitioners who need exact provenance, reproducibility and defensible cross-source correlation.
- Reviewers, auditors and security leads who must challenge a conclusion without depending on the original analyst's memory.
- Small security teams that need a measured workflow on modest hardware rather than a heavyweight SIEM deployment.

The product response: an evidence-reconstruction engine

TraceSOC turns supported raw evidence into canonical observations, detector evaluations, bounded relationships, hypotheses, findings, narratives and response records while preserving a mechanically reversible path back to the original artifact identity.

```
raw evidence
-> representation extraction
-> canonical observation
-> evidence graph
-> detection
-> bounded relationship / correlation
-> hypothesis / finding
-> narrative
-> response
```

The reverse requirement is equally important:

```
downstream conclusion
-> finding
-> relationship / detection
-> canonical observation
-> exact raw locator
-> original artifact
-> SHA-256
```

Six evidence families, twelve frozen representations

The controlled reference demonstrates six evidence families - email, packet, Zeek, Sysmon, PowerShell and Windows Security. Within those families, TraceSOC supports the twelve frozen representations listed in the technical report. That distinction avoids implying a universal parser.

> **No-silent-loss rule:** Unknown or partially supported evidence is preserved and accounted rather than silently discarded or converted into invented semantics.

How the work progressed from lessons to a frozen product

1. Stage 9A supplied reusable engineering lessons - provenance, observation-versus-inference separation, deterministic identity, bounded correlation, falsifiers, clean reproduction and holdout discipline. Case-specific answers and prior runtime were not copied into TraceSOC.
2. Those lessons became a reusable evidence contract: every consequential downstream object had to remain traceable to source evidence, and unsupported/partial representations had to remain visible.
3. TraceSOC expanded into real evidence representations across email, packet, Zeek, Sysmon, PowerShell and Windows Security sources rather than remaining a synthetic fixture exercise.
4. Detection and correlation were built as bounded evidence relationships. Time, byte identity and process identity were never allowed to silently become causality or maliciousness.
5. A genuine analytical inflation issue appeared in the Sysmon-to-PCAP path: 66 raw packet relationships were correct evidence, but would over-inflate downstream reasoning if each became its own final conclusion. The repair preserved all 66 relationships and aggregated only the downstream analytical layer into three source-level hypotheses.
6. The controlled six-source R3 reconstruction established the full pipeline and a complete downstream-to-raw reverse trace.
7. Anti-hardcoding controls and external real Zeek evidence tested whether the system generalized beyond the reference case.
8. The technical candidate was frozen before semantic holdout access. Only then was the sealed post-freeze holdout adjudicated.
9. Finally, clean reproduction, regression, public acceptance and resource measurements were used to prove that the accepted behavior is repeatable and viable on the constrained workstation.

> The engineering story is intentionally not "we got a good detector score." It is "we built a system that keeps evidence, inference and uncertainty defensible across the entire analytical path."

What a judge can independently verify

5,442 / 5,442	524 / 524	11
REFERENCE OBSERVATIONS RAW-LOCATED	**CURRENT REGRESSION PASS**	**CORE HASHES REPRODUCED**
controlled six-source R3	4 / 4 public acceptance	two fresh frozen runs

Controlled six-source reconstruction

- 5,442 canonical observations; all 5,442 retain raw locators.
- 32,658 detector evaluations.
- 5 matched detections.
- 6 hypotheses, 6 findings, 6 narratives and 6 response records after source-level aggregation.
- One complete consequential chain mechanically traced to original artifact SHA-256.

Generalization controls

23,145 / 23,145	156,458 / 156,458	938,748
EXTERNAL ZEEK ROWS RECONCILED	**SEALED HOLDOUT ROWS RECONCILED**	**HOLDOUT DETECTOR EVALUATIONS**
source-count difference 0	post-freeze	0 matched detections

> Holdout detection accuracy remains UNADJUDICATED. Zero holdout detections are not scored as success or failure because no valid exact labeled comparator exists for the sealed input.

Measured workstation behavior

- 198,463 accepted records; 0 quarantined.
- End-to-end runtime: 15.9617856 seconds.
- Peak working set: 486,924,288 bytes against the frozen 4,000,000,000-byte ceiling.
- Two fresh six-source reproductions completed in approximately 3.438 and 3.393 seconds.

How the judge can prove the claims without trusting the prose

1. Open project-control/finalist-candidate-freeze-v04.json and verify the frozen commit and tree were bound before semantic holdout evaluation.
2. Open evidence/reference-case/case-manifest.json and confirm the accepted six-source R3 counts and full raw-locator coverage.
3. Open evidence/final-proof/r3-end-to-end-reverse-trace-proof.json and walk the selected chain from response through finding, hypothesis, detection/relationship and canonical observation to raw locator and original artifact SHA-256.
4. Open the external-evidence proof/ledger entries and verify 23,145/23,145 Zeek rows reconciled without source-count loss.
5. Open evidence/final-proof/holdout-final-adjudication.json and verify 156,458/156,458 holdout rows, 938,748 detector evaluations, no post-freeze tuning, and DETECTION_ACCURACY_STATUS=UNADJUDICATED.
6. Open results/acceptance/public-test-results.json and verify 524 discovered, 524 run, 524 passed and 4/4 public acceptance classes PASS.
7. Open evidence/final-proof/frozen-clean-reproduction-resource-validation.json and verify two fresh frozen runs reproduced the same 11 core hashes as accepted R3.
8. Open project-control/phase4j-cov011-benchmark-result-v1.json and verify the measured 198,463-record resource result.
9. Use finalist-evidence-ledger.json and evidence-index.csv as the navigation layer connecting claims to proof artifacts and their limitations.

> The intended reviewer experience is evidence-first: report claim -> evidence index/ledger -> proof record -> canonical observation/relationship -> exact raw locator -> original artifact identity.

Why the capability matters - and where it deliberately stops

Operational impact

Outcome	What TraceSOC changes	Claim boundary
Clearer investigations	Observed fact, inference, alternatives, confidence and does-not-prove boundaries remain distinguishable.	No universal accuracy claim.
Safer analysis	Correlation does not silently become causality or maliciousness; unsupported evidence remains visible.	No claim that evidence preservation detects every threat.
More resilient handoff	A reviewer can recover exact source evidence and reproduce accepted outputs.	No measured analyst-time reduction.
Stronger auditability	Machine-readable claims bind to proof artifacts, locators and SHA-256 identities.	No monetary ROI claim.
Practical deployment evidence	Measured real-format workload stays within the frozen workstation memory ceiling.	Not a universal scalability guarantee.

Major design defenses

- Provenance is part of the event contract because an analytical conclusion that cannot recover its source evidence is weaker under handoff and challenge.
- Representation-specific adapters are used because different evidence formats have different semantics; a false universal schema would hide important distinctions.
- Unsupported evidence is retained/accounted because silent loss creates false completeness.
- Correlations are deliberately narrow because time adjacency, byte identity and process attribution are corroborative relationships, not automatic causality or maliciousness.
- Packet evidence is preserved at evidence level while downstream reasoning is aggregated, avoiding both data loss and finding inflation.
- The candidate was frozen before holdout evaluation so unseen evidence could not become another development fixture.

Limitations are part of the answer, not footnotes

- Controlled six-source reference activity is not malicious ground truth.
- Correlation does not independently establish causality or maliciousness.
- Post-freeze holdout detection accuracy is UNADJUDICATED.
- Cross-publisher network holdout diversity is not established; development and sealed network corpora are both IoT-23.
- No confirmed-malicious IOC is asserted from the finalist reference evidence; the zero-row IOC set is intentional.
- Response records do not claim executed live production containment.
- Resource measurements are measured-case evidence, not universal scalability claims.
- No monetary savings, analyst-hours saved, false-positive reduction percentage or breach-cost reduction was measured.

> **Future validation:** The strongest next validation is a genuinely independent network publisher/corpus evaluated under the same pre-freeze, no-tuning and provenance-accounting discipline.

Executive conclusion

TraceSOC demonstrates a defensible way to move from heterogeneous raw security evidence to analytical conclusions without losing the evidentiary chain. Its most important result is not a fabricated verdict or score. It is that, within the measured supported boundary, evidence remains recoverable, unsupported certainty remains visible, accepted outputs reproduce deterministically, and a reviewer can independently verify how the conclusion was reached.